Introduction to Machine Learning

Sample course notes for testing a RAG project

1. What is Machine Learning?

Machine learning is a field of artificial intelligence in which computers learn patterns from data instead of being given explicit rules. A machine learning model is trained on examples, and then it makes predictions on new data it has not seen before.

The term was popularised by Arthur Samuel in 1959, who built a checkers program that improved by playing against itself. Today machine learning is used in spam filters, recommendation systems, speech recognition and medical diagnosis.

2. Types of Machine Learning

Supervised learning

In supervised learning the training data contains both inputs and the correct outputs (labels). The model learns to map inputs to outputs. Two main tasks are classification, where the output is a category such as spam or not spam, and regression, where the output is a number such as a house price.

Unsupervised learning

In unsupervised learning the data has no labels. The model looks for structure by itself. Clustering, for example K-Means, groups similar items together. Dimensionality reduction, for example PCA, reduces the number of features while keeping the important information.

Reinforcement learning

In reinforcement learning an agent interacts with an environment and receives rewards or penalties. It learns a policy that maximises the total reward over time. It is used in game playing and robotics.

3. The Machine Learning Workflow

Step 1. Collect the data.

Step 2. Clean the data: remove duplicates, handle missing values and fix errors.

Step 3. Split the data into a training set (usually 80 percent) and a test set (usually 20 percent).

Step 4. Choose a model and train it on the training set.

Step 5. Evaluate the model on the test set using a suitable metric.

Step 6. Tune the hyperparameters and repeat until the results are good enough.

Step 7. Deploy the model and monitor its performance over time.

4. Overfitting and Underfitting

Overfitting happens when a model learns the training data too well, including its noise, so it performs badly on new data. Signs of overfitting are very high training accuracy but low test accuracy. Ways to reduce overfitting include getting more data, using regularisation, using a simpler model, and applying cross-validation.

Underfitting is the opposite problem. The model is too simple to capture the pattern in the data, so it performs badly on both the training set and the test set. A more complex model or better features can help.

The bias-variance tradeoff describes this balance. High bias leads to underfitting and high variance leads to overfitting. The goal is a model with low bias and low variance.

5. Common Algorithms

Algorithm	Type	Typical use
Linear Regression	Supervised (regression)	Predicting house prices
Logistic Regression	Supervised (classification)	Spam detection
Decision Tree	Supervised	Loan approval decisions
K-Nearest Neighbours	Supervised	Simple classification
K-Means	Unsupervised (clustering)	Customer segmentation
PCA	Unsupervised	Reducing the number of features

6. Evaluation Metrics

Accuracy is the fraction of predictions that are correct. It can be misleading when one class is much more common than the other.

Precision is the fraction of predicted positives that are truly positive. **Recall** is the fraction of actual positives that the model finds. The **F1 score** is the harmonic mean of precision and recall.

For regression, common metrics are Mean Absolute Error (MAE), Mean Squared Error (MSE) and R-squared.

Practice questions

• What is the difference between supervised and unsupervised learning?

• Name two ways to reduce overfitting.

• Why can accuracy be misleading?

• What is the usual train/test split mentioned in these notes?